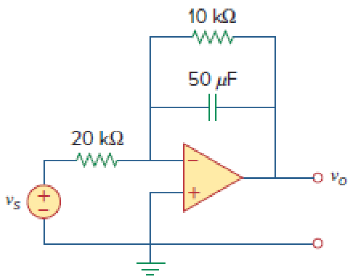
Chapter 16, Problem 66P

Problem

For the op amp circuit in Fig. 16.89, find $v_0(t)$ for $t > 0$. Take $v_s = 12\, e^{-5t}\, u(t)$ V.

Figure 16.89:



Step-by-step solution

Step 1 of 2

Refer to Figure 16.89 in the text book.

Determine the feedback impedance of the circuit.

$$
\begin{aligned}
Z_F(s) &= \left(\frac{1}{sC}\right) \| R \\
&= \frac{\left(\frac{1}{sC}\right) R}{\frac{1}{sC} + R} \\
&= \frac{R}{1 + sCR}
\end{aligned}
$$

Substitute $10\ \text{k}\Omega$ for R and $50\ \mu\text{F}$ for C.

$$
\begin{aligned}
Z_F(s) &= \frac{10000}{1 + s(10000)(50 \times 10^{-6})} \\
&= \frac{10000}{1 + 0.5s} \\
&= \frac{20000}{s + 2}
\end{aligned}
$$

Step 2 of 2

Express the source voltage in Laplace domain.

$$V_s(s) = \frac{12}{s+5}$$

Determine the output voltage expression.

$$\begin{aligned}
V_o(s) &= -\frac{Z_F(s)}{20\ \text{k}\Omega} V_s(s) \\
&= -\left(\frac{\frac{20000}{s+2}}{20000}\right)\left(\frac{12}{s+5}\right) \\
&= -\left(\frac{12}{(s+2)(s+5)}\right) \\
&= \frac{4}{s+5} - \frac{4}{s+2}
\end{aligned}$$

Apply inverse Laplace transform.

$$v_O(t) = 4\left(e^{-5t} - e^{-2t}\right)u(t)$$

Thus, the expression of $v_O(t)$ is $\boxed{4\left(e^{-5t} - e^{-2t}\right)u(t)}$.